

AI-Based Smart Healthcare Emergency Monitoring System

Healthcare institutions such as hospitals, intensive care units (ICUs), emergency wards, and elderly care centers continuously face challenges in monitoring patient conditions and responding quickly to medical emergencies. Traditional patient monitoring systems mainly depend on manual observation by doctors and nurses, which becomes difficult when handling a large number of patients.

In many situations, patients may suddenly experience critical health conditions such as falls, breathing difficulties, unconsciousness, emotional distress, seizures, or abnormal movements. Delayed identification of these emergencies can increase health risks and affect patient safety. Continuous monitoring of every patient also increases the workload on healthcare staff and may lead to human errors during emergency situations.

The proposed project aims to develop an intelligent healthcare monitoring platform capable of analyzing multiple forms of patient data such as images, videos, audio signals, and medical text records to detect emergency situations automatically in real time. The system analyzes patient facial expressions, ICU surveillance videos, breathing or cough audio, distress voice recordings, and clinical text reports to identify abnormal patient behavior and critical medical conditions.

By combining information from multiple data sources, the system can generate emergency alerts and assist healthcare professionals in taking faster and more accurate decisions. The project focuses on improving patient safety, reducing manual monitoring workload, and supporting efficient healthcare management through an AI-powered monitoring solution.

Dataset Description:

The dataset consists of two categories: **Normal Frames** and **Affected Frames**.

- The **Affected Frames** folder contains healthcare monitoring images capturing patients in critical or abnormal conditions such as falls, seizures, unconsciousness, breathing difficulty, distress behavior, or other emergency situations. These frames are annotated with corresponding JSON files that define the location and boundaries of affected patients or abnormal regions of interest.
- The **Normal Frames** folder contains routine patient monitoring images where no emergency or abnormal condition is observed. These frames represent stable patient conditions and do not require any annotations.

Dataset Link:

https://drive.google.com/drive/folders/1CCRMwmmM5UsM5fEi7GywbzPjQHfA_XRIL?usp=drive_link